Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **ZEUS SHIELD** (project: SENTINEL AI)

b. Team leader name: **Mohammed Ilyas**

c. Problem Statement: Scammers can drain a customer's savings in minutes, before a bank can react. Singapore's rapid account-draining safeguard calls for a hold before settlement, but fraud teams need a transparent, auditable way to enforce it, explain it and act on it in real time. SENTINEL AI is that control tower: rules decide, AI explains, humans authorize.

Brief about the idea

Detect. Explain. Intervene. Investigate.

SENTINEL AI is a real-time fraud operations control tower for retail banking. It turns Singapore's rapid account-draining safeguard into a deterministic policy gate.

For a personal account with a protected balance of S$50,000, a transfer that, with the previous 24 hours of outflows, moves more than 50% of the balance is held before it settles.

A pre-transaction decision API returns ALLOW, HOLD or REJECT with an itemised risk breakdown. Bank operators review each hold in a live Control Tower, Alerts & Cases and an investigation workspace.

The operating principle

Rules decide

Seven pure rules and one policy gate set every outcome.

AI explains

Gemini explains a decision already made. It cannot change a score.

Humans authorize

Release needs customer verification and is audited.

Opportunities

1 How different is it from existing ideas?

- Most fraud tools score risk with opaque models. SENTINEL AI enforces an explicit, versioned policy gate.
- The gate acts before settlement, not after the money has left.
- Built around one concrete safeguard: the S$50,000 protected balance and 50% in 24 hours rule.

2 How will it solve the problem?

- Holds a draining transfer in the decision API, before it settles.
- Shows operators an itemised score and the evidence behind it.
- Contacts the customer, verifies, then releases or blocks, with every step on an append-only audit trail.

3 USP of the proposed solution

- Rules decide. AI explains. Humans authorize.
- Gemini output is schema-validated and every claim must cite a field in the evidence bundle, or it is removed.
- Explanations never recommend releasing a hold.

List of features offered by the solution

1. **Transaction simulator**
   Deterministic takeover scenario on synthetic data, plus seeded noise traffic.

2. **Real-time risk engine**
   Seven pure rules and a policy gate return ALLOW, HOLD or REJECT.

3. **Rapid-drain detection**
   50% of balance in 24 hours, with configurable thresholds.

4. **Intervention workflow**
   Customer contact, account kill-switch and verified release, all audited.

5. **Investigation workspace**
   Deterministic attribution, timeline and evidence bundle per case.

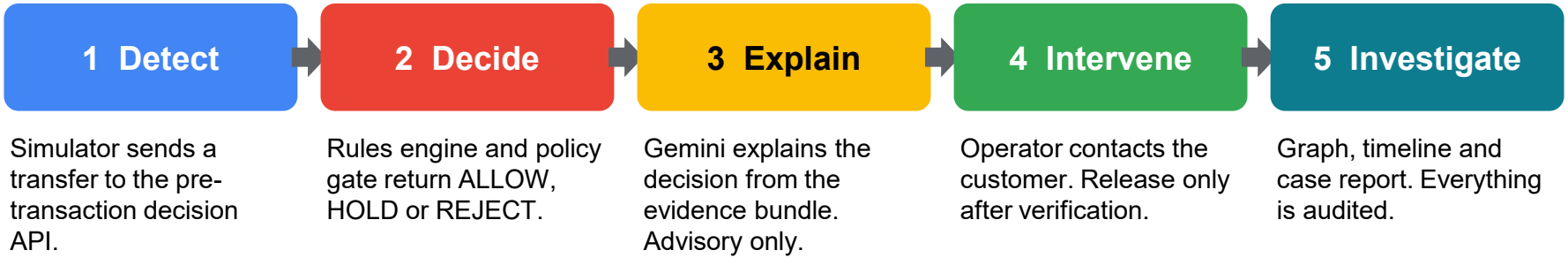
6. **AI explanation and summary**
   Gemini on Vertex AI, advisory only, every claim cites evidence.

7. **Entity graph**
   Server-side two-hop network and a fund trace that stops at the observed boundary.

8. **Case reports and audit**
   Report lifecycle, analyst sign-off, PDF export and a full audit log.

Also: Control Tower, Live Transactions, Accounts, Policies & Rules, Analytics, Integrations, bank data lens, performance panel.

Process flow diagram

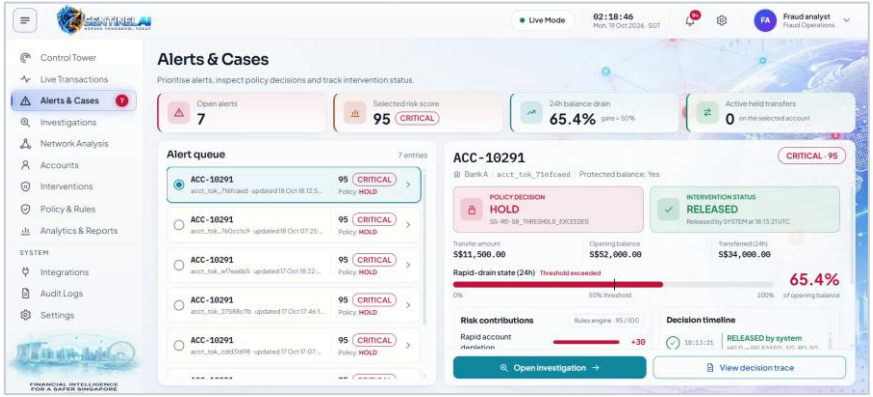


Worked example from the demo: account takeover on ACC-10291

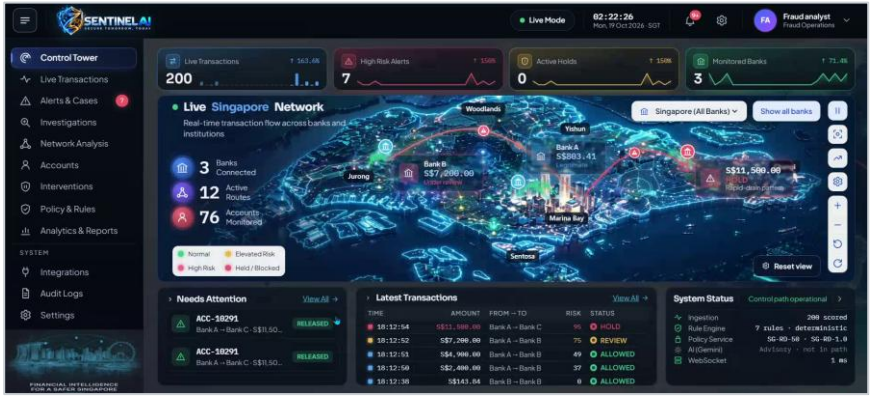
S$52,000	S$34,000	65.4%	95
opening balance	moved in 24 hours	of balance, gate is 50%	CRITICAL, policy: HOLD

Rule SG-RD-50 fires, the transfer is held before settlement, and the operator releases it only after customer verification.

Wireframes/Mock diagrams of the proposed solution



Alerts & Cases (light theme)

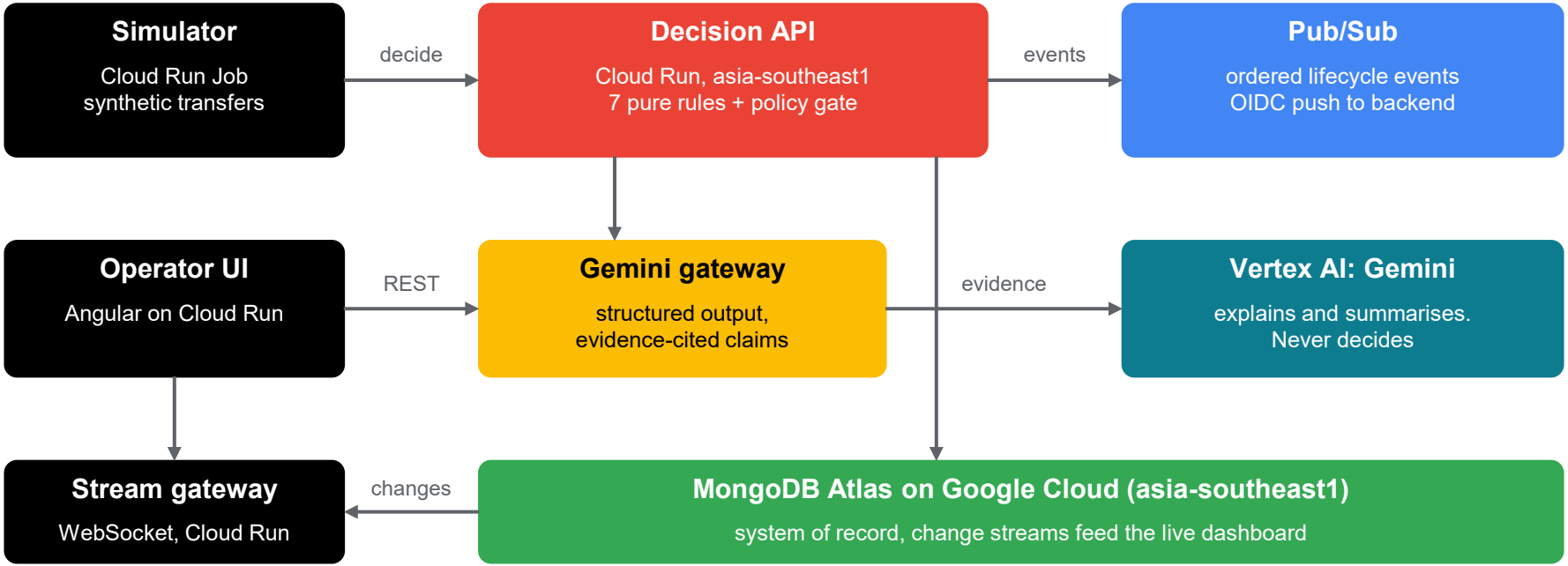


Control Tower (dark theme)

Design language: a single console for bank operators with light and dark themes, a persistent module navigator, and the policy decision always shown next to AI output.

These are screens of the working prototype, not static mock-ups.

Architecture diagram of the proposed solution



Cloud Build + Artifact Registry deploy to Cloud Run on every push to main. Secret Manager holds the Atlas connection string. No API keys in the repo.

Technologies to be used in the solution

Google Cloud

- Vertex AI: Gemini, structured output
- Cloud Run: UI, decision API, stream gateway, simulator job
- Pub/Sub: ordered lifecycle events
- Cloud Build and Artifact Registry
- Secret Manager

Application

- Angular for the operator console
- TypeScript workspaces: domain, rules, scenarios, core, stream, simulator, ui
- WebSocket live stream
- ESLint import-boundary rules

Data and quality

- MongoDB Atlas on Google Cloud, change streams
- Bounded two-hop graph traversal
- 405 automated tests
- Synthetic data only

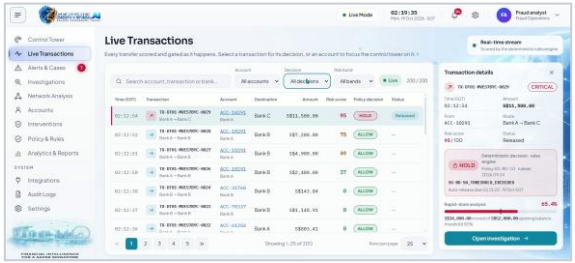
Gemini runs only from the backend with the service account's own credentials. Model IDs are configuration, not code, and the browser never sees one.

Estimated implementation cost (optional)

Component	Cost model	How it is kept low
Cloud Run (4 components)	Per request and CPU time	Scale to zero. Simulator runs as a job.
Vertex AI: Gemini	Per token	Cached per assessment. Demo account pre-warmed. Never in the decision path.
Pub/Sub	Per volume delivered	Lifecycle events only. The decision never waits on it.
MongoDB Atlas	Cluster tier	Single deployment on synthetic data for the demo.
Cloud Build, Artifact Registry, Secret Manager	Small usage-based fees	One secret. One build per push to main.

A priced estimate is not included because the prototype has not been load-tested in a deployed environment. The design choices above are the main cost controls.

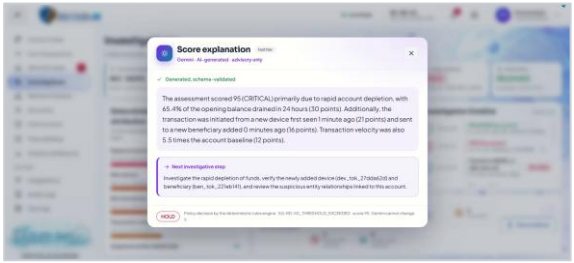
Snapshots of the prototype (1 of 2)



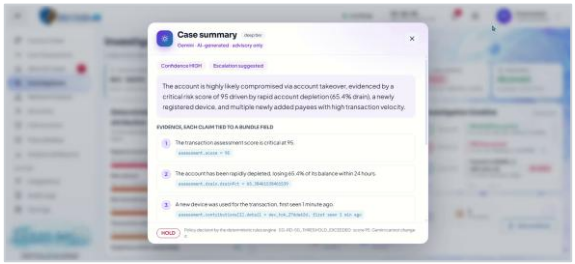
Live Transactions with policy decision



Investigation workspace



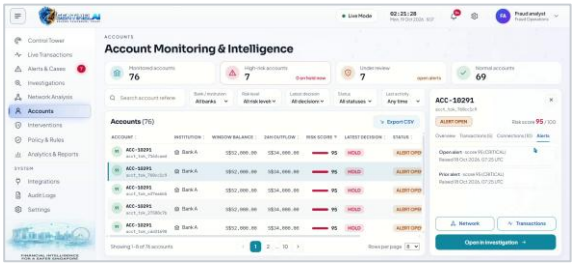
Gemini score explanation



Gemini case summary with evidence

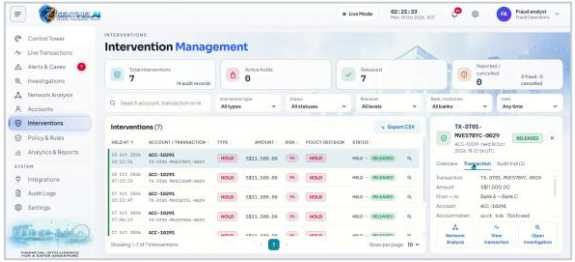


Transaction & entity network

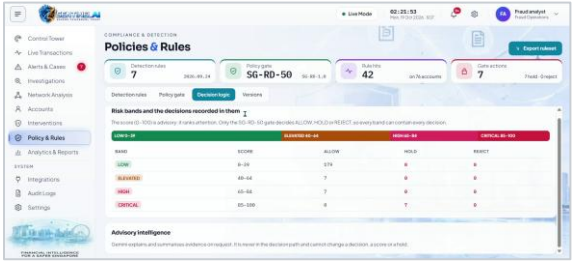


Account monitoring & intelligence

Snapshots of the prototype (2 of 2)



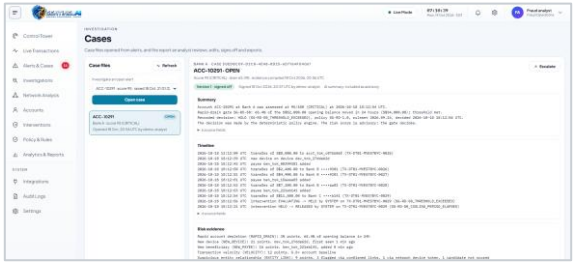
Intervention management



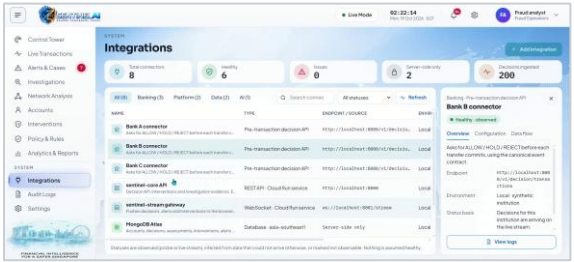
Policies & Rules: policy gate



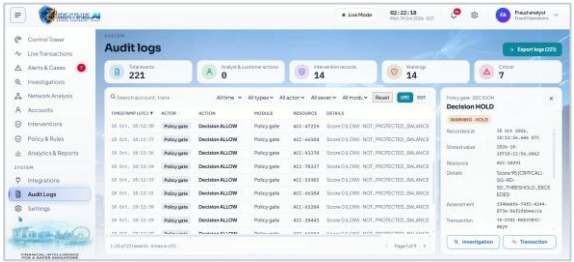
Analytics & Reports



Case report with sign-off



Integrations



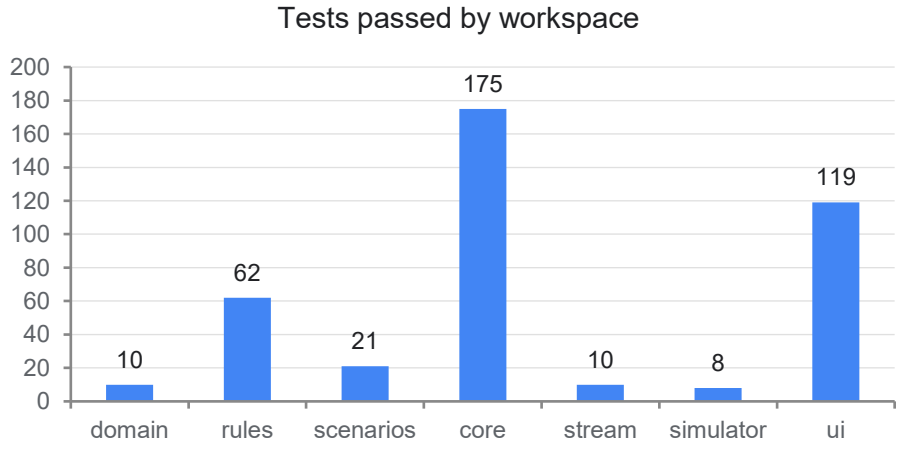
Audit logs

Prototype Performance report/Benchmarking

405
automated tests passed

22
control-path scenario tests

0
language-model calls in the decision path



What was verified

- npm run verify passed on 19 Oct 2026: typecheck, lint, tests and builds for all workspaces
- Takeover repeats identically; seeded noise scores ELEVATED and never reaches HOLD
- Cross-bank reads are refused (403) and audited before any model call
- Checked live: signed case report to PDF, fund trace, kill switch, bank lens

Measured locally, not yet deployed

Laptop to remote Atlas: decision API p95 about 1.0 s and graph query p95 about 0.6 s, above the 80 ms and 150 ms targets. Deployed and Gemini timings are not yet measured.

Additional Details/Future Development

Per-tenant sign-in

The bank lens is enforced by the API and the stream today; next, derive it from a per-tenant identity provider.

Scenario launcher in the cloud

Built for local runs and the Cloud Run Job; next, operator sign-in and the job invoker permission.

Browser end-to-end coverage

Automated browser tests for case reports, lens switching, themes and responsive sizes.

Meet latency targets

The performance panel now measures; next, cache graph queries and benchmark in region.

Deployment and Gemini verification

Load and latency checks in an authorized cloud environment.

Gemini outage drill

A control that shows holds and audit continuing while the model is unavailable.

Scope note: all institutions and data are synthetic. No real bank or COSMIC integration is claimed.

Provide links to your:

GitHub Public Repository

https://github.com/Mohammed-Ilyas-AI/Sential-AI

Demo Video Link (3 Minutes)

Add the video link here once it is uploaded

Final Product Link

https://sentinal-ai.cloud.run/

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

